

Thermal Analysis of CubeSat Solar Panels

Abstract

This paper presents a thermal analysis of solar panel configurations for 3U CubeSat missions in low Earth orbit. We evaluate three panel geometries using finite element methods and validate predictions against on-orbit telemetry from the TechSat-1 mission.

1. Introduction

Small satellites in low Earth orbit experience significant thermal cycling between sunlight and eclipse. Solar panels must withstand temperature swings from -65 C to +125 C during each 90-minute orbital period.

Previous studies focused on large spacecraft thermal control. CubeSat missions impose unique constraints including limited surface area, restricted mass budgets, and minimal active thermal control options.

Figure 1. Temperature distribution across the panel surface.

2. Methodology

The thermal model was constructed using a finite element approach with 2400 shell elements representing the solar panel substrate. Boundary conditions included solar flux during sunlight phases and radiative cooling to deep space during eclipse. Material properties were taken from manufacturer datasheets for the Spectrolab UTJ solar cells used on TechSat-1.

3. Results

Configuration	Peak Temp (C)	Delta-T (C)
Body-mounted	128.3	193.3
Single-deploy	114.7	179.7
Double-deploy	110.2	175.2

The double-deploy configuration achieves the lowest peak temperature of 110.2 C, representing a 14.1 percent reduction compared to body-mounted panels. The increased radiating surface area of deployable panels enables more efficient heat rejection.

4. Conclusions

This study demonstrates that deployable solar panel configurations offer significant thermal advantages for 3U CubeSat missions in LEO. The validated finite element model agrees with TechSat-1 telemetry to within 3.2 C RMS error. Future work will extend the analysis to highly elliptical orbits.